

02 — Ohm's law, dividers, power and energy

Class notes p.1, p.4–5, set 2 p.4–6 · MTE syllabus item 2 · 4 board questions · needs 01

Map

1 Voltage division

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2 Current division

→

3 Division with AC

→

4 Power and energy

→

5 Ladder reduction

The questions this file answers

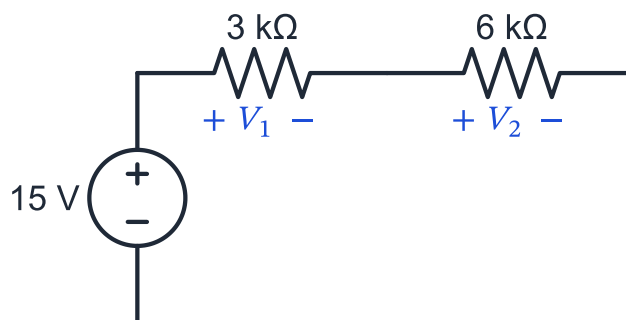
- **Class p.4** 15 V across 3 kΩ and 6 kΩ in series. Find V_1 and V_2 .
- **Class p.5** A 12 A source feeds 3 kΩ || 6 kΩ || 4 kΩ. Find the current in each.
- **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 kΩ and 7 kΩ in series. Find v_1 and v_2 .
- **Class p.1** An infinite ladder: 3 kΩ shunt, 5 kΩ series, 7 kΩ shunt, 11 kΩ series, 13 kΩ shunt, ... Find R_{eq} .

Build

1 · Voltage division

CLASS P.4

15 V is applied across 3 kΩ and 6 kΩ in series. Find the voltage across each.



Series resistors carry the same current, so each one's voltage is proportional to its resistance.

GUESS FIRST

Which resistor gets more voltage, and roughly how much?

Series means **one current**: $I = \frac{V}{R_1 + R_2}$. Each voltage is that current times its own resistance:

$$V_1 = V \cdot \frac{R_1}{R_1 + R_2} \quad V_2 = V \cdot \frac{R_2}{R_1 + R_2}$$

$V_1 = 15 \times \frac{3}{9} = 5 \text{ V}$, $V_2 = 15 \times \frac{6}{9} = 10 \text{ V}$. They add back to 15 V (KVL).

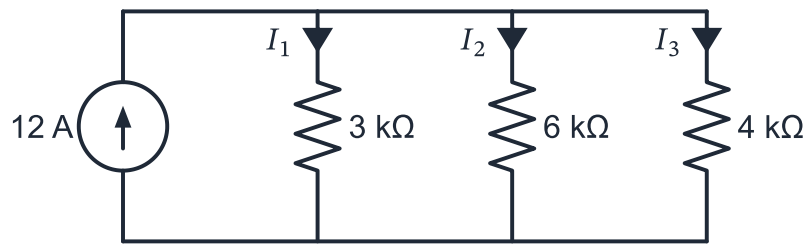
CHECK 1

12 V across 1 k Ω , 2 k Ω , 3 k Ω in series. Voltage across the 2 k Ω ?

2 · Current division

CLASS P.5

A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .



Parallel branches share one voltage, so the current splits in inverse proportion to resistance.

GUESS FIRST

Which branch carries the most current? Rank the three before calculating.

Parallel means **one voltage**. For two branches that gives the class rule, where **each branch gets the other resistor on top**:

$$I_1 = I \cdot \frac{R_2}{R_1 + R_2} \quad I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

For three branches, the professor's method is to merge the other two branches into one, then apply the two-branch rule:

Branch	The other two, merged	Current
3 k Ω	$6 \parallel 4 = 2.4 \text{ k}\Omega$	$12 \times \frac{2.4}{2.4 + 3} = 5.33 \text{ A}$
6 k Ω	$3 \parallel 4 = \frac{12}{7} \text{ k}\Omega$	$12 \times \frac{12/7}{12/7 + 6} = 2.67 \text{ A}$
4 k Ω	$3 \parallel 6 = 2 \text{ k}\Omega$	$12 \times \frac{2}{2 + 4} = 4 \text{ A}$

Check: $5.33 + 2.67 + 4 = 12 \text{ A}$ (KCL). Faster in one line: the current in each branch is proportional to its **conductance** $1/R$, i.e. to $1/3 : 1/6 : 1/4 = 4 : 2 : 3$, so 12 A splits as $12 \times \frac{4}{9}$, $12 \times \frac{2}{9}$, $12 \times \frac{3}{9}$.

LABEL IN THE CLASS NOTES

The side note " $I_2 + I_3 = 6 \times 4 / 10 = 2.4$ " is the **resistance** $6 \parallel 4 = 2.4 \text{ k}\Omega$, not a current.

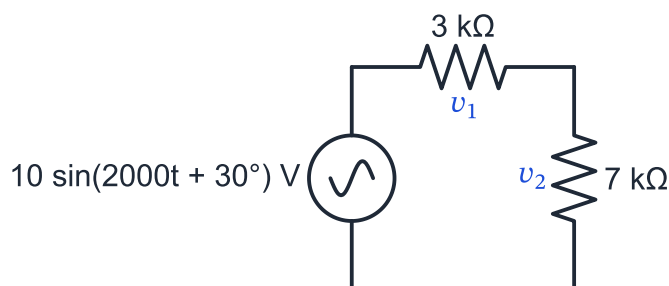
CHECK 2

(a) 10 mA into $1 \text{ k}\Omega \parallel 4 \text{ k}\Omega$: current in the $1 \text{ k}\Omega$? (b) Why does the *smaller* resistor take the *larger* current?

3 • Division works for AC too

CLASS P.5

$v = 10 \sin(2000t + 30^\circ) \text{ V}$ is applied across $3 \text{ k}\Omega$ and $7 \text{ k}\Omega$ in series. Find v_1 and v_2 .

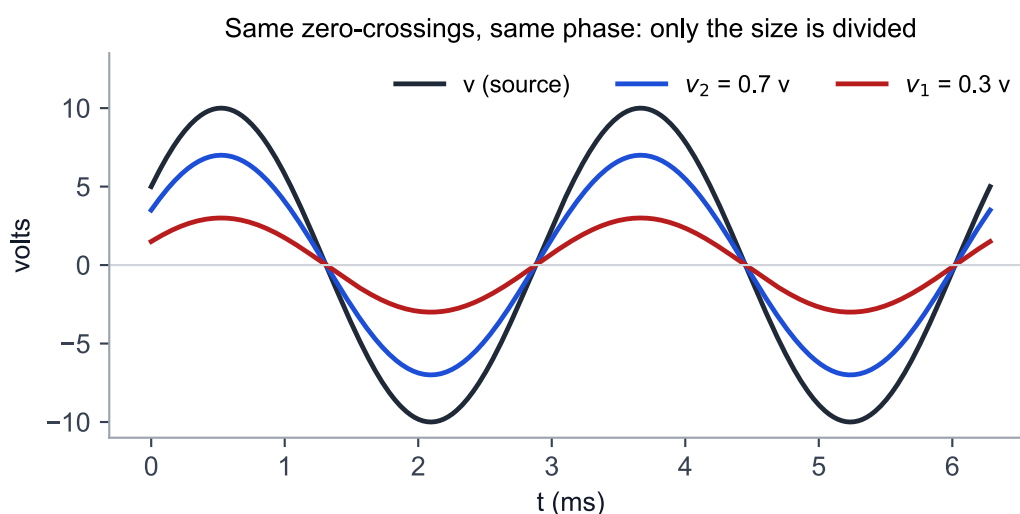


GUESS FIRST

Does v_1 have the same phase angle as the source, or a different one?

A resistor's $v = iR$ holds **at every instant**, so the divider ratio applies to the whole waveform. Resistors do not shift phase:

$$v_1 = 0.3 \times 10 \sin(2000t + 30^\circ) = 3 \sin(2000t + 30^\circ) \text{ V} \quad v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$$



All three waves cross zero together: same frequency, same phase. Only the amplitude is divided.

This holds only because both elements are **resistors**. With an L or C in the divider, the ratio becomes complex and shifts the phase. That case uses impedances (05 §6, 06 §6).

4 • Power and energy: the sign convention

PREDICT

A 12 V battery drives 1 mA. Another circuit forces 1 mA *into* the battery's + terminal (charging it). In which case does the battery absorb power?

Passive sign convention. Label each element's voltage + / -. If the current arrow **enters the + terminal**, then

$$p = v \cdot i \quad \text{is the power ABSORBED by the element}$$

- $p > 0$: the element **absorbs** power (a resistor, always; a battery being charged).
- $p < 0$: the element **delivers** power (a source driving a circuit).
- If the current **leaves** the + terminal, the absorbed power is $p = -v \cdot i$.

Units: $V \times mA = mW$, so $12 V \times 1 mA = 12 mW$. For the Predict: current entering + $\Rightarrow$ the battery absorbs 12 mW (charging); current leaving + $\Rightarrow$ it delivers 12 mW.

Energy is power over time:

$$w = \int p \, dt \quad (\text{joules; constant power: } w = P \cdot t)$$

Element	Power absorbed	Energy
R	$p = i^2 R = \frac{v^2}{R} \geq 0$	all turned into heat
L	$p = Li \cdot \frac{di}{dt}$	stored: $\frac{1}{2} Li^2$
C	$p = Cv \cdot \frac{dv}{dt}$	stored: $\frac{1}{2} Cv^2$

Conservation (Tellegen): in any circuit the absorbed powers add to zero. What the sources deliver, the other elements absorb. 03 uses this as the check on every loop and node answer.

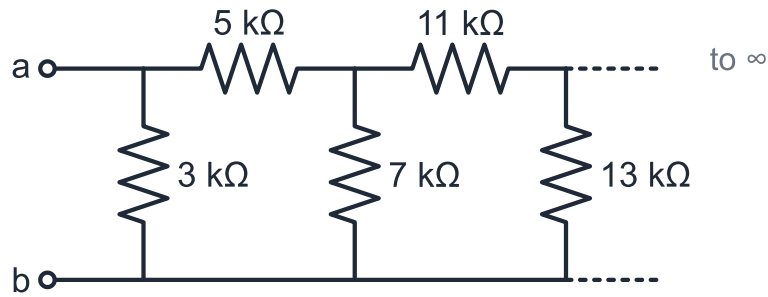
CHECK 3

A 5 V source has 2 mA **leaving** its + terminal. Absorbed power? Is it delivering or absorbing?

5 • Series–parallel reduction: the infinite ladder

CLASS P.1 • THE FIRST QUESTION OF THE COURSE

Find R_{eq} of the infinite ladder: 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ... ∞ .



Read from the far end: each shunt sits in parallel with everything to its right; each series resistor adds.

GUESS FIRST

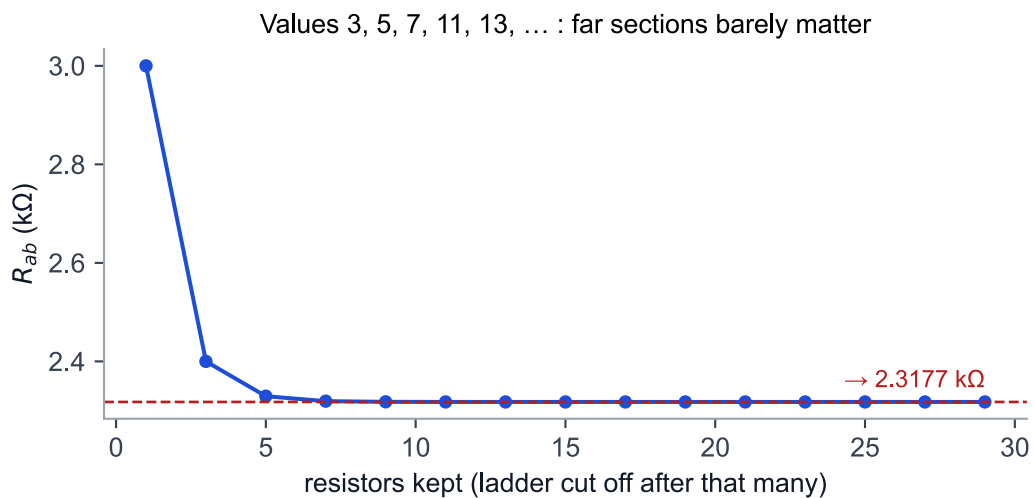
The first resistor is 3 kΩ, directly across a–b. Can R_{eq} be more than 3 kΩ?

Rule: reduce from the far end towards the terminals. Written out, the ladder is:

$$R_{eq} = 3 \parallel (5 + 7 \parallel (11 + 13 \parallel (...))) \quad \text{k}\Omega$$

The Guess answer: the 3 kΩ is in parallel with everything else, so $R_{eq} < 3 \text{ k}\Omega$, always.

The values 3, 5, 7, 11, 13 are the **prime numbers**. Assuming that pattern continues (the notes stop at 13), cut the ladder off after more and more resistors:



3 resistors give 2.400 kΩ, 5 give 2.329, 9 give 2.318. The far sections are huge resistors hidden behind parallel paths, so they barely change the answer.

$R_{eq} \approx 2.318 \text{ k}\Omega$ (converged; confirmed by computer). The professor may intend a different continuation, so ask.

THE STANDARD TRICK, AND WHY IT FAILS HERE

For a ladder whose sections are **all identical** (series R, shunt R, repeated forever), cutting off the first section leaves the **same** infinite ladder. So $R_{eq} = R + (R \parallel R_{eq})$. Solving the quadratic gives $R_{eq} = R(1 + \sqrt{5})/2 \approx 1.618 R$.

Here every section is different, so the rest of the ladder never looks like the whole, and you must truncate.

Exam form

- **VDR:** $V_k = V \cdot \frac{R_k}{R_{total}}$. **CDR (two branches):** $I_1 = I \cdot \frac{R_2}{R_1 + R_2}$.
- Division holds for any waveform when every element is a resistor. Phase is unchanged.
- **Power:** $p = vi$ absorbed when i enters $+$. $p > 0$ absorbs, $p < 0$ delivers. Energy $w = \int p \, dt$; $\frac{1}{2}Li^2$, $\frac{1}{2}Cv^2$.
- **Ladder:** reduce from the far end. Identical infinite ladder: $R_{eq} = R + R \parallel R_{eq}$.

Traps

- CDR with the **same** resistor on top ($I \cdot R_1 / (R_1 + R_2)$ for I_1) is the most common slip: the answer comes out swapped.
- Power sign: a source whose current **leaves** $+$ delivers power, so its absorbed power is **negative**.
- Forgetting that R_{eq} of a network with a shunt resistor across the terminals must be smaller than that resistor.

Self-test

1. 20 V across 2 k Ω and 8 k Ω in series. Both voltages?
2. 9 mA into 3 k $\Omega \parallel$ 6 k Ω . Both currents?
3. $v = 6 \sin(100t)$ V across 1 k Ω and 2 k Ω in series. v across the 2 k Ω ?
4. 3 mA flows through a 2 k Ω resistor for 10 s. Power, and energy turned into heat?
5. An infinite ladder of identical 1 k Ω sections (series then shunt). R_{eq} ?

Answers

Check 1. $12 \times 2/6 = 4 \text{ V}$.

Check 2. (a) $10 \times 4/5 = 8 \text{ mA}$. (b) Same voltage across both, and $I = V/R$, so the smaller R takes more current.

Check 3. $p = -5 \times 2 = -10 \text{ mW}$ absorbed: it **delivers** 10 mW.

Self-test.

1. **4 V** and **16 V**.

2. $3 \text{ k}\Omega$: $9 \times 6/9 = 6 \text{ mA}$; $6 \text{ k}\Omega$: **3 mA**.

3. $\frac{2}{3} \times 6 \sin(100t) = 4 \sin(100t) \text{ V}$.

4. $P = (3 \text{ mA})^2 \times 2 \text{ k}\Omega = 18 \text{ mW}$; $w = 18 \text{ mW} \times 10 \text{ s} = 0.18 \text{ J}$.

5. $R(1 + \sqrt{5})/2 = 1.618 \text{ k}\Omega$.